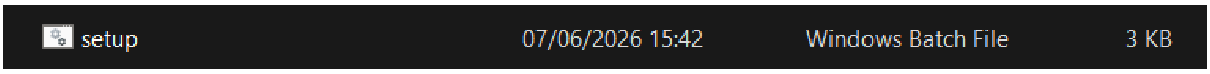
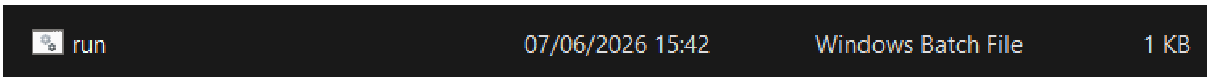
Running the First Test

A. Installation and Setup

a. Run setup.bat (or setup.sh) to install the required dependencies into a local virtual environment.

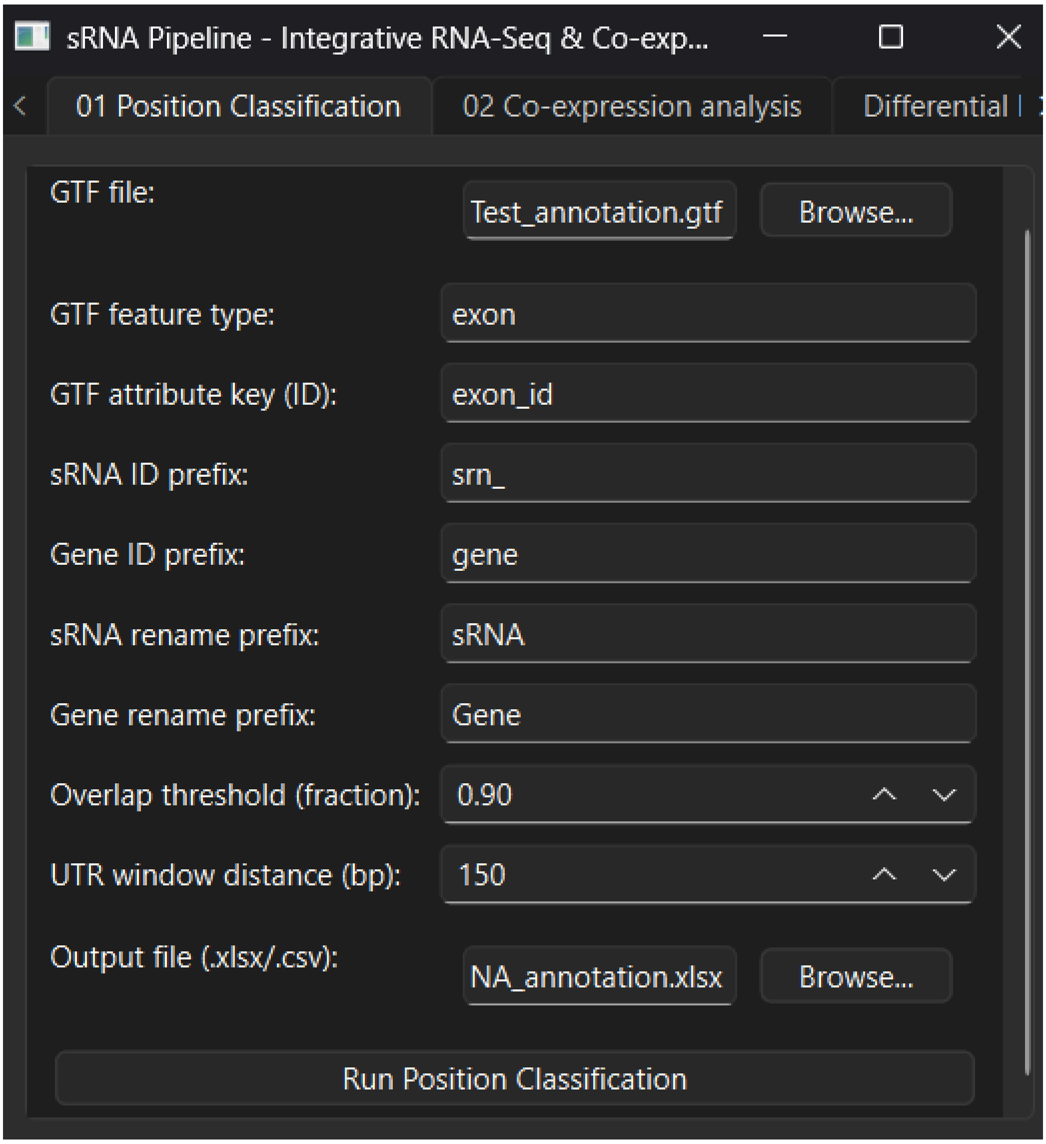


B. Once installation has completed, run run.bat (or run.sh) to launch the application.



C. Optionally, create a dedicated folder in which to store the results generated throughout this tutorial.

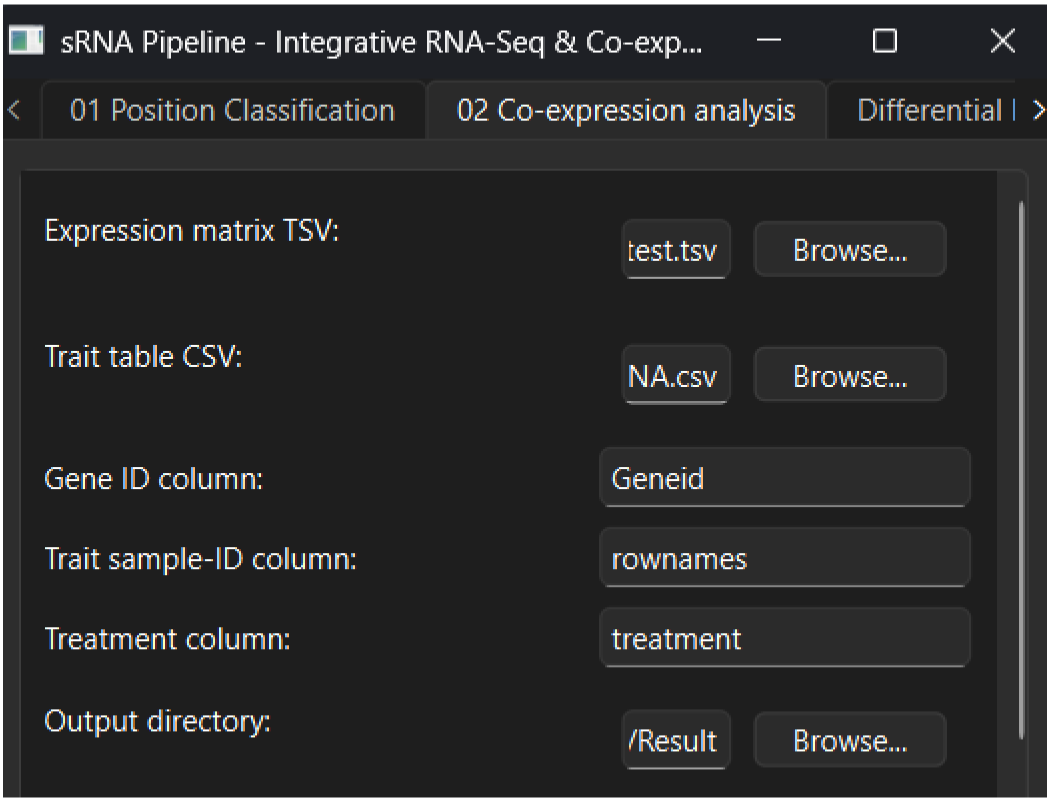
1. Position Classification



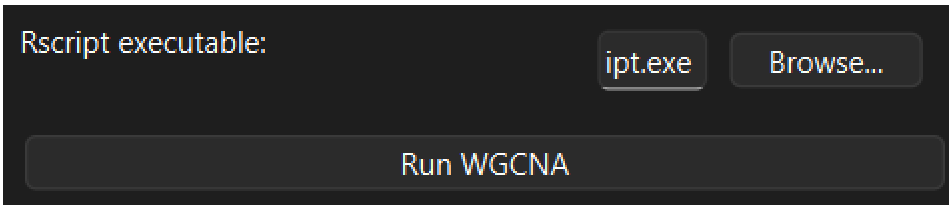
a. Select the GTF annotation file Test_annotation.gtf, located in the 00 Test data folder.
b. Specify the desired output file.
c. Click Run Position Classification.
d. **Expected output:**

```
Found 681 sRNAs and 3013 genes.
Category counts:
location
Intergenic 374
Antisense 219
3' UTR 31
5' UTR 30
Intragenic 27
Saved sRNA_annotation.xlsx in the desired output directory.
```

2. Co-expression Analysis (WGCNA)



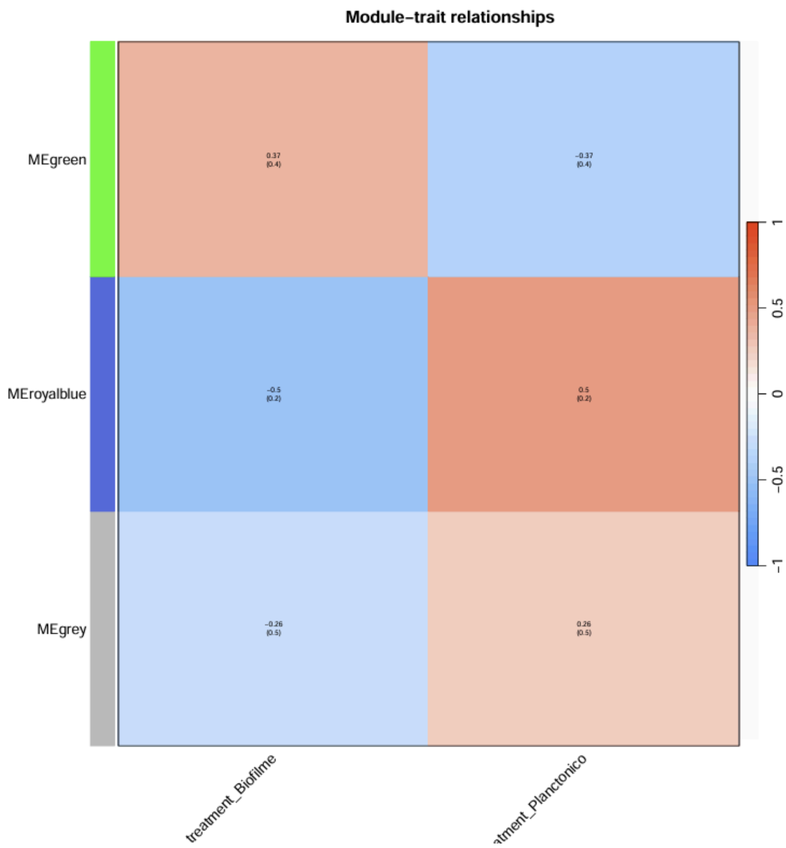
Input: select Count_test.tsv as the expression matrix (TSV).
Select table_treatment_WGCNA.csv as the trait table (CSV).
Set the output directory.
Adjust the RStudio (Rscript) executable path if required.
Click Run WGCNA.



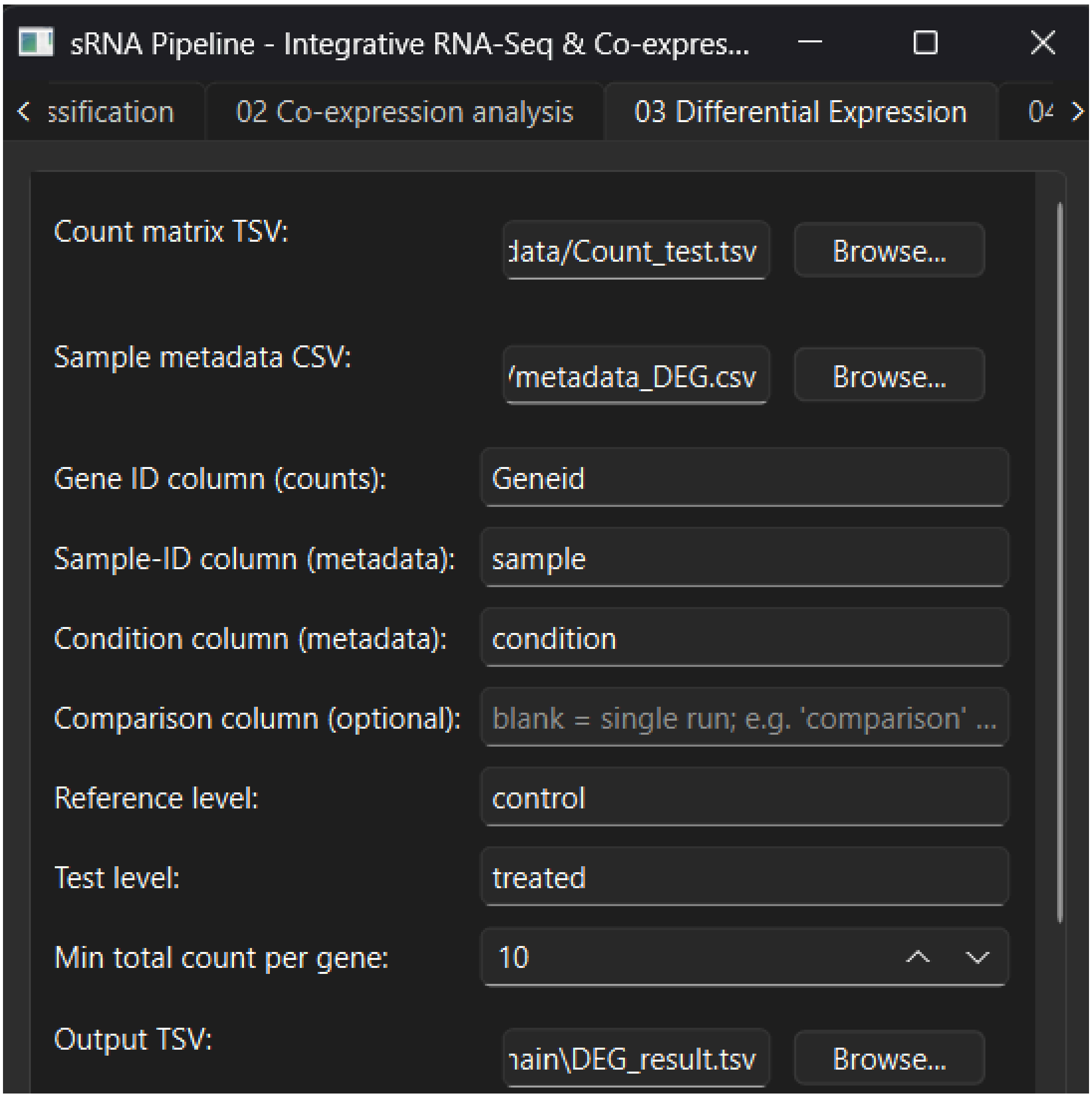
The following files are generated in the results folder:

top_hub_genes_per_module	10/06/2026 13:30	Comma Separated...	1 KB
gene_module_membership	10/06/2026 13:30	Comma Separated...	74 KB
CytoscapeInput-edges	10/06/2026 13:30	Text Document	133,722 KB
CytoscapeInput-nodes	10/06/2026 13:30	Text Document	71 KB
gene_dendrogram_with_modules	10/06/2026 13:30	PNG File	153 KB
Module_Trait_Heatmap	10/06/2026 13:30	Microsoft Edge PD...	6 KB
soft_thresholding_power_selection	10/06/2026 13:30	PNG File	61 KB
sample_dendrogram_with_traits	10/06/2026 13:30	PNG File	63 KB
sample_dendrogram_outlier_check	10/06/2026 13:30	PNG File	54 KB
WGCNA_configured	10/06/2026 13:30	R Source File	8 KB

A module-trait heatmap is also produced for comparison:



3. Differential Expression (DESeq2)

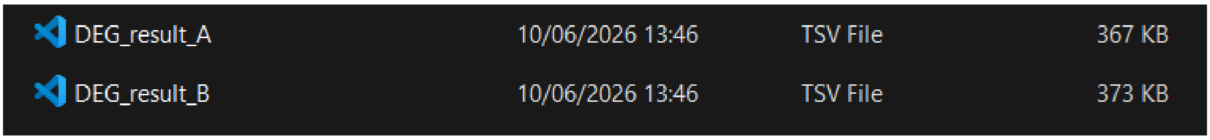


Input: set the count matrix TSV to Count_test.tsv.
Set the sample metadata CSV to metadata_DEG.csv.
For this test, set the Comparison column to comparison. This splits the metadata into groups (e.g., A, B, C, D) and runs DESeq2 once per group, producing one DEG file per comparison (DEG_result_A.tsv, DEG_result_B.tsv, ...).

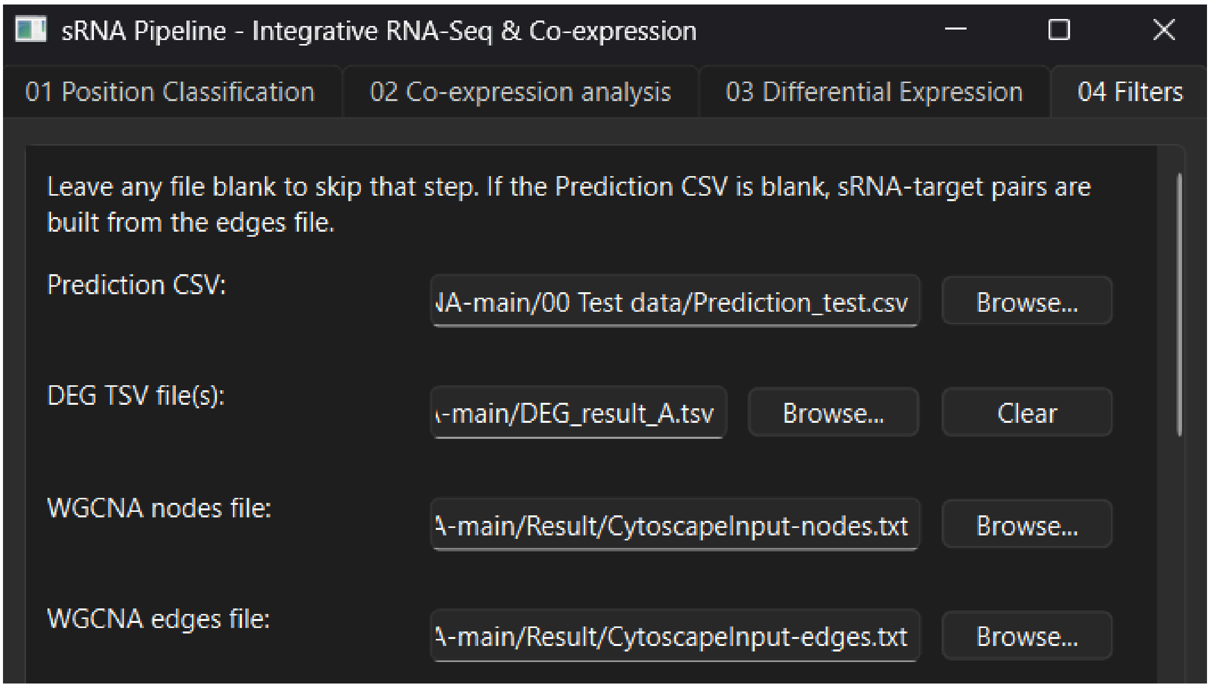
Set the output directory.
Adjust the RStudio (Rscript) executable path if required.
Click Run DESeq2.

Expected output:

```
Shared dispersion estimated from 8 samples over 3211 genes.
comparison A -> DEG_result_A.tsv [own dispersion] sig padj<0.05: 591
comparison B -> DEG_result_B.tsv [own dispersion] sig padj<0.05: 1804
```



4. Filtering



Prediction CSV: select Prediction_test.csv.
Add both DEG_result_A.tsv and DEG_result_B.tsv to the DEG TSV files field.
Add CytoscapeInput-nodes.txt to the Cytoscape Node field.
Add CytoscapeInput-edges.txt to the Cytoscape Edges field.

In the DEG filter configuration, set the minimum strains to 2.

Expected output:

```
Total predictions: 33996
After energy/probability filter (>= 5/5 of ['E_intaRNA', 'E_Rnaplex', 'E_TargetRNA3',
'Probability_TargetRNA3', 'Probability_sRNARFTarget']): 29662
Consistent DEGs: 572
After DEG filter: 423
After module filter (same): 234
After requiring network edge: 183
After best_per_target selection: 146
Final interactions: 146
```